

Atomic Mixing in Metals Under Shear Deformation

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The fundamental processes of shear-induced chemical mixing in heterogeneous Cu-based alloy systems have been studied by molecular dynamics computer simulations. These simulations reveal that two very disparate mechanisms operate depending on whether or not the two phases are coherent. For the coherent systems, mixing occurs as dislocations transfer across phase boundaries. The mixing in these systems is “superdiffusive,” and for spherical precipitates, the rate of mixing increases quadratically with precipitate radius. In systems that have incoherent phases, the mixing occurs by a local shuffling of atoms at the interface, and for them, the mixing is diffusive, with the mixing rates of spherical precipitates scaling linearly with particle radius. The morphologies of the interfaces for the two situations are also different. Coherent precipitates form rough interfaces that are relatively sharp, whereas the interfaces of incoherent precipitates are smooth but diffuse. These simulations also show that for incoherent precipitates, shear-induced mixing can be very different at different crystallographic interfacial planes as well as for different strain directions.

INTRODUCTION

Nanostructured alloys and composites have become a central focus in the search for materials that can withstand the extreme environments anticipated for future energy technologies, hydrogen storage, nuclear energy, shape-memory alloys, motors, and batteries. In many of these situations, top-down processing schemes, such as severe plastic deformation (SPD)—ball milling, accumulative roll bonding (ARB), equal-channel angular extrusion (ECAE), wire drawing or high pressure torsion—are particularly attractive since large quantities of material can be economically processed.¹ While these methods are all known to refine grain sizes to the nanometer length scale² the details of the resulting microstructures can be quite distinct, as these methods impose different deformation modes, ranging from simple shear for high-pressure torsion and ECAE to plane strain for ARB, thus stabilizing different crystallographic textures and possibly different phases. As a consequence, the design of new materials with desired properties would benefit

greatly from an improved fundamental understanding of SPD processing. Also of great importance to alloy design is the phase stability of nanocrystalline alloys under SPD. Many two-phase alloys, such as Cu-Ag³ and Cu-Fe,^{4,5} are known to become single-phase, homogenous alloys under SPD at suitably low temperatures. Similarly, several intermetallic compounds can chemically disorder during SPD and even turn amorphous.⁶ This behavior has been broadly rationalized by the concept that atoms relocate during SPD by “ballistic” mixing processes,⁷ in analogy to the behavior of alloys under energetic particle irradiation.^{8,9} Within this framework, most binary systems should homogenize during SPD, unless the temperature is sufficiently high for thermally activated diffusion processes to contribute, usually considered the temperature where vacancies become mobile. Several experimental studies have shown, however, that this is not the case and that such systems as Ni-Ag,¹⁰ Cu-Ta,¹¹ Cu-Nb,¹² and Cu-Mo¹² remain two-phase during low-temperature SPD, although they do homogenize over length scale as small as

≈ 50 nm. While a detailed understanding of these various behaviors is not currently available, a great deal has been learned about the mechanisms of shear mixing from molecular dynamics (MD) simulations. One of the important findings from these investigations is that the primary mechanism for shear mixing varies from one system to the next, and in fact, it even varies at different interfaces within a single system. In the current article, we illustrate these different mechanisms using a series of Cu-based alloys, and we highlight in a single system, Cu-Nb, the different nature of shear mixing across different interfaces. Details of the MD simulation methods and interatomic potentials can be found in the cited references.

SHEAR MIXING IN fcc-fcc AND fcc-bcc BINARY SYSTEMS

We begin our discussion of atomic mixing during SPD by treating the case of coherent, face-centered cubic (fcc) precipitates in fcc Cu. We characterize the rate of shear mixing by defining the mean *relative* shear-induced diffusion coefficient¹³

$$D(R) = \frac{1}{6} \frac{\partial \sigma^2(R, t_1, t_2)}{\partial \varepsilon} = \frac{1}{6} \frac{\partial}{\partial \varepsilon} \left(\sum_{i=1}^N \sum_{j=1}^{z(R,t)} \left[(R_{ij})_{t_2} - (R_{ij})_{t_1} \right]^2 \right) / \left(2 \sum_{i=1}^N z(R, i) \right) \quad (1)$$

where R_{ij} is the vector distance between atom pairs i and j at times t_1 and t_2 , and Z is the number of such pairs. $D(R)$ represents the movement of atoms relative to other atoms separated by a distance R . The mean relative diffusion coefficient provides a good description of mixing during SPD because it provides information about the local changes in atomic configurations. The first important result of the MD simulations is that $D(R)$ increases linearly with pair separation in coherent systems during uniaxial compression. This is shown in Fig. 1.¹³ (The curvature observed in this figure i.e., $D(R) \propto R(L-R)$ dependence is an artifact of the periodic boundary conditions.) The reason for this rather surprising linear dependence is simply that a dislocation must pass between two atoms to affect their relative motion, and the probability for such an event is proportional to their distance apart when the deformation is homogeneous. In polycrystalline alloys, $D(R)$ increases linearly with R over the length scale of coherent dislocation glide, typically the grain size. Such mixing is “superdiffusive,” as the root mean square separation distance between pairs of atoms grows superlinearly with time,¹³ as shown inset in Fig. 1.

The correlated motion of atoms during shearing just described has interesting consequences for the morphology of the evolving sheared system as well as for the rates of atomic mixing. We have measured

the shear mixing rate at interfaces by counting the number of new A–B bonds per unit strain, $dN_{AB}/d\varepsilon$, where A and B refer to the two components of a binary system.¹⁴ This quantity depends on the initial morphology of the precipitates and the strain path, but for simplicity, we use a spherical precipitate B, of radius R , embedded in matrix A and apply cyclic (biaxial) compressive straining. By the end of a three-step cycle, the shape of the cubic computational cell is restored: $\varepsilon_1 = 0.1\varepsilon_x + 0.1\varepsilon_y$; $\varepsilon_2 = 0.1\varepsilon_y + 0.1\varepsilon_z$; $\varepsilon_3 = 0.1\varepsilon_z + 0.1\varepsilon_x$. The results of the MD simulations are shown in Fig. 2; here, the mixing rate, $Dn_{AB}/d\varepsilon$, is plotted for a number of binary Cu based alloys as a function of the number of atoms in the

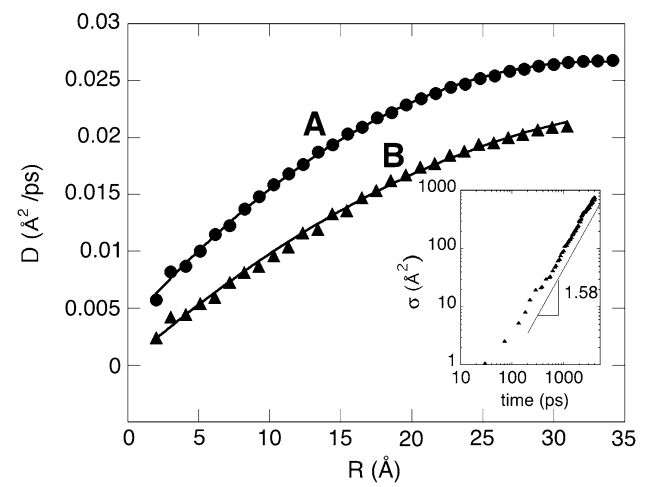


Fig. 1. Apparent diffusion coefficients calculated from Eq. 1 as a function of the separation distance between pairs of atoms, for two runs. The softer matrix in run A leads to higher values of D . The solid lines are best fit of the data with the functional dependence $R(L-R)$. Inset shows that the mean relative square displacement varies as $\sigma^2 \propto (\text{time})^{1.58}$ (after Ref. 13).

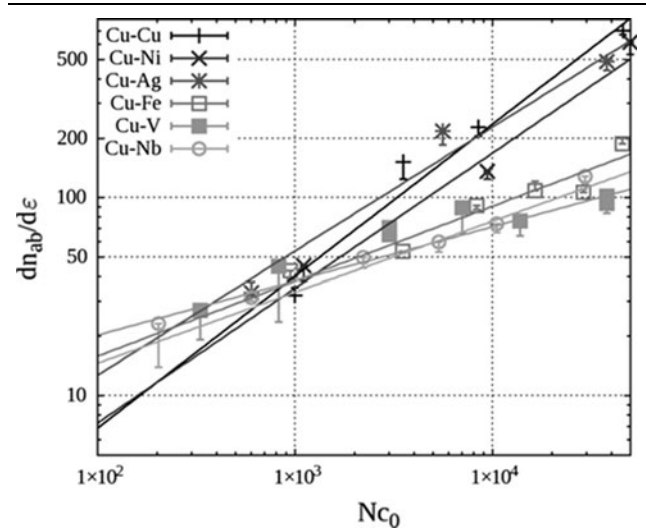


Fig. 2. Rate of A–B bond formation as a function of particle size. The lines are logarithmic fits to the data. Dark gray, fcc-fcc systems; light gray, fcc-bcc systems (after Ref. 14)

sphere using log-log scales. ($\text{Cu}_A\text{-Cu}_B$ refers to Cu precipitates [called B atoms] in a Cu matrix [called A atoms].)

For the coherent precipitates, Cu_B , Ni, and Ag, the mixing scales as R^2 and not as R , as would be expected for dissolution of the precipitate by Fickian diffusion. The R^2 dependence is readily understood by noting that the probability for a dislocation to intersect the spherical precipitate is proportional to R , and a dislocation cutting through the sphere creates a line of A-B bonds at the circumference, which is also proportional to R . The number of new A-B bonds thus scales as R^2 . The change in the precipitate morphology is also unusual. For this case, it is more revealing to show the difference between shear mixing and diffusional mixing at a planar interface, as illustrated in Fig. 3. These results derive from kinetic Monte Carlo simulations,¹⁵ but similar results are found by MD. For a planar (100) interface, it is seen that the interface initially acquires a long-wavelength roughness but otherwise remains rather sharp. New A-B bonds are largely associated at low strain with the creation of new interfacial area rather than intermixing; i.e., long wavelength Fourier components dominate the interface power spectrum.¹⁵ Eventually, shorter wavelength Fourier components grow and the system homogenizes. For diffusional mixing the interface becomes diffuse but does not roughen, as illustrated at the bottom of Fig. 3. The morphologies of spherical precipitates after a strain of six are shown in the top row of Fig. 4 for these same systems.

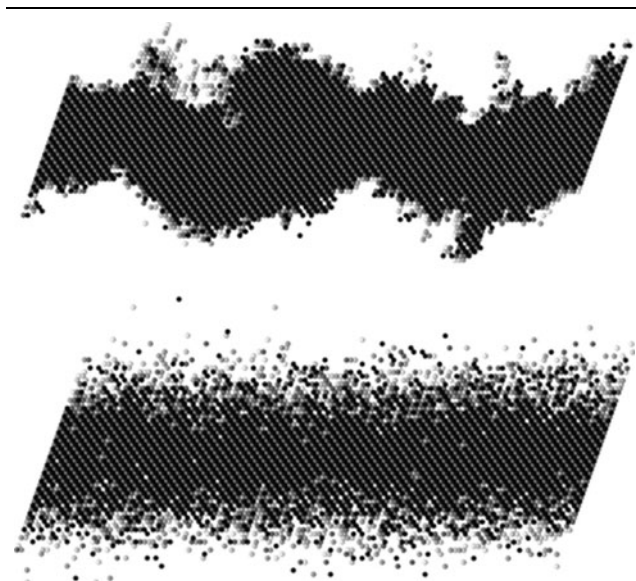


Fig. 3. Cross-sectional views of atoms, subjected to shear mixing (top) or diffusional mixing (bottom). Shear mixing leads to predominantly low-frequency roughening of the interfaces compared to the atomic scale disorder introduced by diffusional mixing. Gray scale indicates distance from the cut plane.

Body-centered cubic (bcc) precipitates are not coherent in Cu, and consequently they behave far differently than the fcc precipitates under shear. Dislocations do not transfer across phase boundaries in these cases, and as a consequence, these particles remain equiaxed and do not develop the long wavelength roughness observed in the fcc systems. This is illustrated in the bottom row of Fig. 4 for V and Nb precipitates in Cu, where the particles are seen to undergo very little deformation during shear and retain their equiaxed morphology. This morphology persists even at strains up to 60.¹⁴ Nevertheless, these particles still undergo shear mixing in Cu but by a different mechanism. The exact atomic mixing process is complex, but it is clear that dislocations pile up at Cu-particle interfaces and create high levels of stress and both chemical and structural disorder. We thus picture the mixing process to proceed as illustrated in the schematic energy diagram in Fig. 5. Mixing occurs by mechanically lowering energy barriers in the disordered region, which leads to local mechanical instabilities and atom exchanges. The heat of mixing and the relative atomic size of the alloys system also play roles in this process since they influence the local energy barriers that must be overcome during relaxation. Vo et al.,¹⁶ for example, showed in fcc metals with very high heats of mixing, that such local atomic rearrangements around dislocation cores can lead to precipitation of initially homogeneous alloys during SPD, even during shearing at 100 K. The mixing mechanism, in fact, resembles that in amorphous materials where shearing forces local relaxations without any long-range correlations in the atomic motion. For this reason, the shear mixing is diffusive rather than superdiffusive, as revealed by the linear mixing rate $dN_{AB}/d\varepsilon$ for Cu-Nb and Cu-V plotted in Fig. 2. We note that Malandro and Lacks¹⁷ used this same picture several years earlier to explain the mechanical response of amorphous alloys to mechanical shear.

SHEAR MIXING AT VARIOUS Cu-Nb INTERFACES

We turn next to considering shear mixing at different interfaces within a given system. We choose as examples three heterointerfaces with Kurdjumov-Sachs (KS) and Nishiyama-Wassermann (NW) orientation relationships (ORs) in the Cu-Nb system. For the KS OR, two distinct habit planes were studied: $\{111\}_{\text{FCC}} \parallel \{110\}_{\text{BCC}}$, which we refer to as $\{111\}\text{KS}$, and $\{112\}_{\text{fcc}} \parallel \{112\}_{\text{bcc}}$, which we refer to as $\{112\}\text{KS}$ (Table I). These interfaces have all been observed experimentally. The NW and $\{111\}\text{KS}$ interfaces are found in Cu-Nb multilayers prepared by physical vapor deposition,^{18,19} whereas the $\{112\}\text{KS}$ is the dominant interface in samples prepared by accumulative roll bonding.²⁰⁻²²

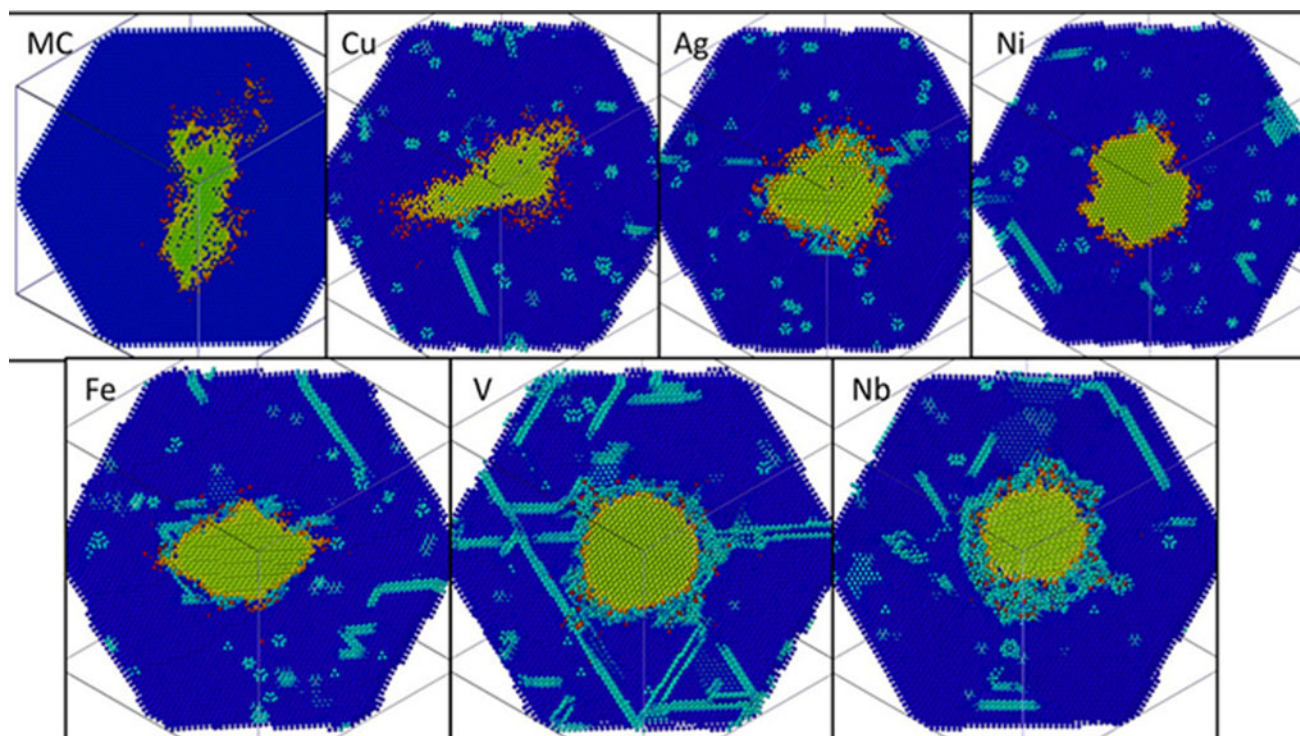


Fig. 4. Morphology of spherical particles in Cu after a strain of 6.0. The left-most configuration is from a MC simulation of a Cu precipitate in a Cu matrix. The rest of the configurations are from MD simulations of precipitates in a Cu matrix. The particle material is indicated in each panel. Color indicates the structural factors for the matrix atoms; blue are fcc atoms and gray are atoms positioned in a locally non-pure fcc environment (e.g., a stacking fault or next to a point defect). For the particle atoms color indicates local mixing; green indicates that most of the neighbors are from the particle and red indicates that most atoms are from the matrix (after Ref. 14). (Color version of figure is available online).

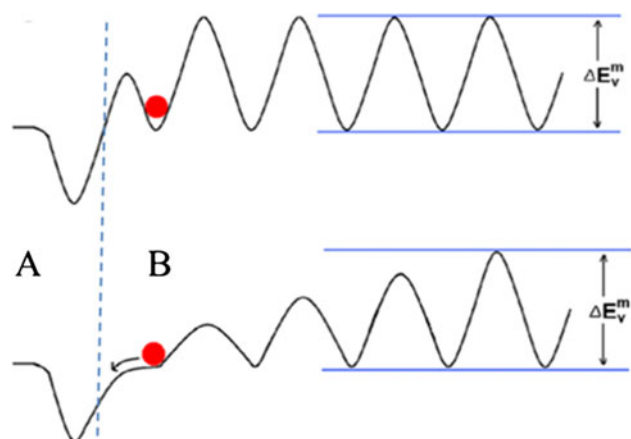


Fig. 5. Energy landscape seen by a B atom near a disordered A-B interface (top). In the presence of applied stress (bottom), some barriers are lowered enabling atomic exchanges.

After relaxation for 30 ps, the {111}KS and NW interfaces are seen in Fig. 6 to remain flat and atomically sharp. In contrast, Shockley partial misfit dislocations are formed above the {112}KS interface with the dislocation lines lying along the y -axis, i.e., $[110]_{\text{Cu}}$. These partials are found in the Cu layer and are connected to the interface plane by intrinsic stacking faults (SFs), which correspond to two layers of atoms of hexagonal close-packed (hcp)

structure. These interface structures are in good agreement with those reported previously for this system.^{23–27}

These three types of Cu/Nb interfaces offer significantly different resistance to shear deformation, as illustrated in Fig. 7 where the flow stress is plotted as a function of in-plane strain direction α .²⁸ For all of these interfaces, the flow stress is highly anisotropic. The NW interface exhibits a two-fold symmetry around $\alpha = 0^\circ$, i.e., the $[011]_{\text{Cu}}$ direction. This is a symmetry element of the NW interface. A similar two-fold symmetry is found for the flow stress plots for the {112}KS interface, about a 0° to 180° direction (i.e., $[11\bar{1}]_{\text{Cu}}$), and this symmetry can also be directly attributed to the symmetry of the {112}KS interface. In the case of the {111}KS interface, there is only an approximate symmetry about a 150° to 330° axis. This is consistent with the less symmetric structure of that interface. We note that a different directional dependence of interface shear strength (ISS) was reported for the {111}KS by Wang and coworkers^{24–27}; however, these authors employed bicrystals instead of multilayers used here. Significantly, the magnitude of the flow stress for the NW and {111}KS along any shearing direction, and for the {112}KS along $\alpha = 90^\circ/270^\circ$, i.e., the $[110]_{\text{Cu}}$ direction, is much smaller than those for {112}KS along other directions. Direct visualization of the atomic configurations during straining

Table I. Orientations and dimensions of the Cu/Nb interfaces

Interface	Orientations						Dimensions			
	Cu (fcc)			Nb (bcc)			X (nm)	Y (nm)	Layer thickness (nm)	
	x-axis	y-axis	z-axis	x-axis	y-axis	z-axis			Cu	Nb
NW	$[01\bar{1}]$	$[\bar{2}11]$	$[111]$	$[001]$	$[\bar{1}\bar{1}0]$	$[110]$	10.0	8.9	5.7	5.1
{111}KS	$[01\bar{1}]$	$[\bar{2}11]$	$[111]$	$[\bar{1}11]$	$[\bar{2}\bar{1}1]$	$[011]$	9.7	8.1	5.7	5.1
{112}KS	$[11\bar{1}]$	$[\bar{1}10]$	$[112]$	$[\bar{1}10]$	$[\bar{1}\bar{1}1]$	$[112]$	9.4	9.7	6.2	4.9

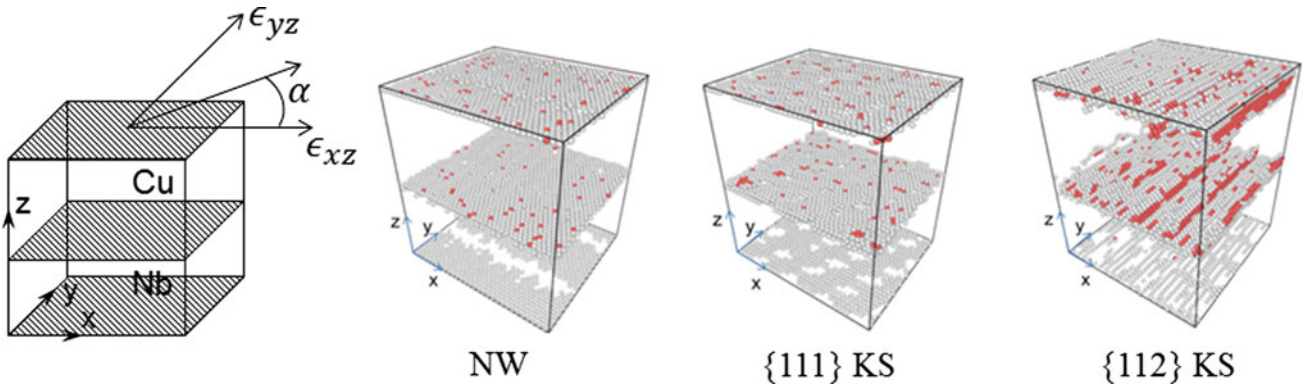


Fig. 6. Schematic of interface geometry. The angle α denotes the shearing direction. Also given are the initial configurations after relaxation for NW, {111}KS, and {112}KS, where the atoms of hcp and disordered local environments are colored in red and grey, respectively (for the colors, refer to the online version). The atoms with fcc and bcc environments are not shown for clarity.

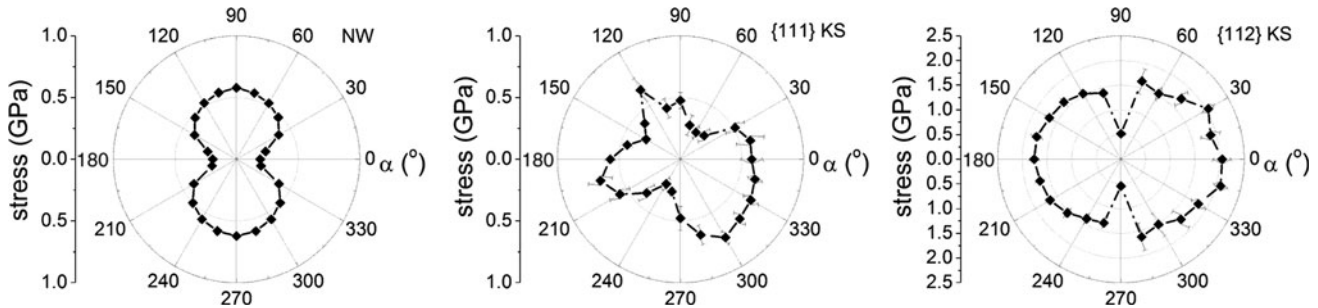


Fig. 7. Dependence of flow stress on shear direction for three Cu/Nb interfaces (after Ref. 28).

reveals, in fact, that in the former cases the samples deformed by pure interfacial sliding, while in the latter some dislocation activity occurred in the Cu phase.

Shear Mixing at {111}KS and NW Interfaces

Shear mixing during simple shear deformation is found to be closely related to the interface shear resistance. In particular, interfaces with low ISS are generally able to deform by pure sliding, and no mixing takes place. This is illustrated in Fig. 8 for the case of the NW interface sheared along $\alpha = 45^\circ$. The sliding mechanism for the {111}KS interface was reported to be based on the motion of in-plane misfit

dislocations.^{24–27} Similar mechanisms are expected to be valid for the NW interface. In both cases, no dislocation activity is observed within the Cu or Nb layers, and the interface remains atomically sharp for the largest strains investigated, $\varepsilon = 20$.

Shear Mixing at {112}KS Interfaces

In contrast to the cases of NW and {111}KS interfaces, the microstructural evolution of {112}KS interfaces subjected to simple shear deformation varies significantly with the shear direction, as illustrated in Fig. 9, and consequently, so does shear mixing (Fig. 10). For example, for the case of small strains and $\alpha = 90^\circ/270^\circ$, deformation is fully

accommodated by interface sliding, similar to the NW and {111}KS interfaces. Consequently, there is little or no mixing at small strains (see Fig. 10). At

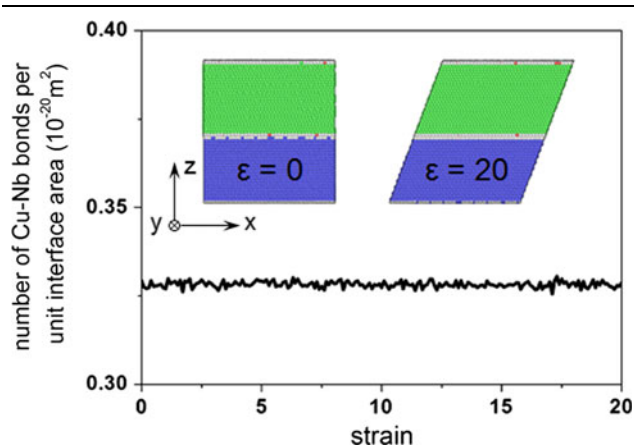


Fig. 8. Number of Cu-Nb bonds per unit interface area as a function of strain for the NW interface sheared along $\alpha = 45^\circ$. Snapshots of atomic configurations at strain $\varepsilon = 0$ and 20 are shown in insets. The atoms are colored based on their local environments: green, blue, red and white for fcc, bcc, hexagonal close-packed (hcp), and disorder, respectively (for the colors, refer to the online version).

larger strains, on the other hand, a structurally disordered layer forms at the original Cu/Nb interfaces, and mixing becomes significant. A detailed analysis shows that this layer is amorphous. By using a standard common neighbor analysis, we find that this layer grows approximately as the square root of strain (Fig. 11). This suggests a diffusive (and not superdiffusive) mixing mechanism. We point out that amorphization at Cu-Nb interfaces has been observed in both experiments^{29,30} and simulations.^{14,31}

We next turn to the case of $\alpha = 0^\circ$ and 180° . At small strain, ε less than ≈ 0.2 , distinct shearing directions yield distinct deformation behaviors, as reported by Demkowicz and Thilly.²³ Shear along the positive ($\alpha = 0^\circ$) or negative ($\alpha = 180^\circ$) x -axis causes the initial partials to extend across the Cu layer, or to be reabsorbed by the interface, respectively (compare Fig. 9b1, f). Upon further shearing for $\alpha = 0^\circ$, more SFs as well as microtwins are created in the Cu layer (Fig. 9b2). At larger strains, amorphous layers form, and the deformation is then almost entirely accommodated by the localized shearing of these layers. A similar localization of plastic deformation in the amorphous layers is

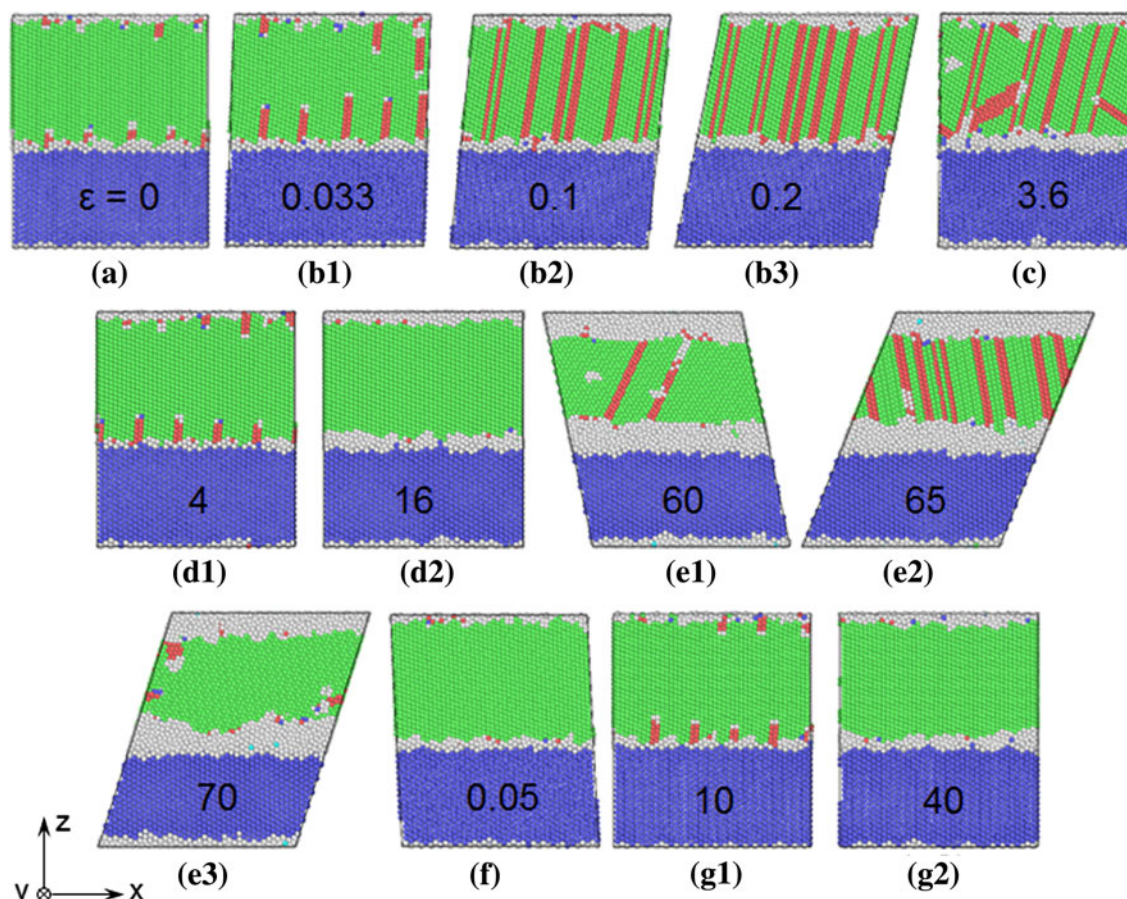


Fig. 9. Snapshots for the {112}KS interface subjected to shearing. The initial state (a) and the strained states for α equal to (b1–3) 0° (c) 45° , (d1–2) 90° , (e1–3) 135° (f) 180° , and (g1–2) 270° . The strain magnitudes are denoted. The atoms are colored based on their local environments: green, blue, red, and white for fcc, bcc, hcp, and disorder, respectively (for the colors, refer to the online version).

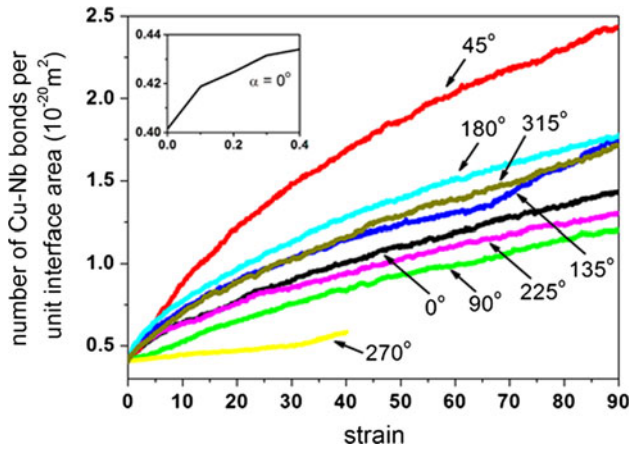


Fig. 10. Number of Cu–Nb bonds per unit interface area as a function of shear strain for the {112}KS interface. Each curve is labeled with its corresponding shearing direction.

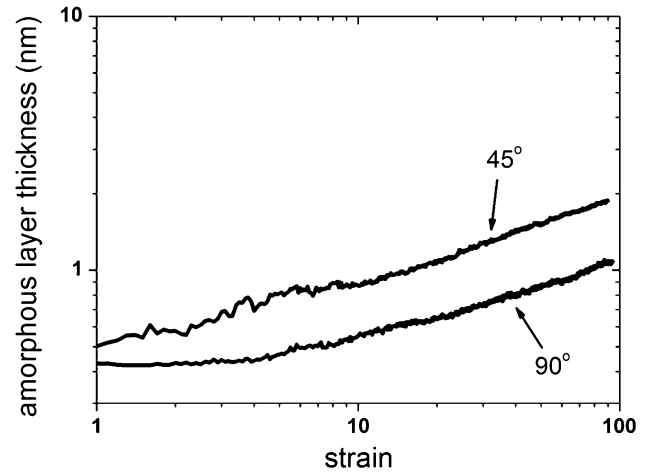


Fig. 11. Thickness of the disordered (amorphous) interfacial layer as a function of strain for $\alpha = 45^\circ$ and $\alpha = 90^\circ$ for the {112}KS interface.

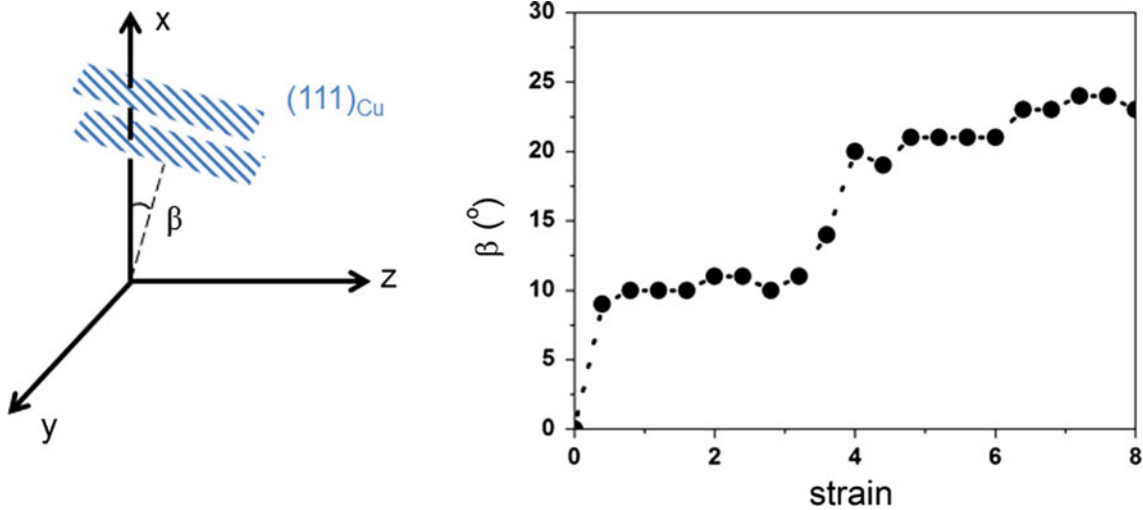


Fig. 12. Orientation change of (111)_{Cu} for the {112}KS interface with $\alpha = 45^\circ$.

observed for all shearing directions for the {112}KS interface.

Another interesting observation is that the crystallographic orientation of the Cu phase does not remain constant during deformation. An example is shown in Fig. 12 for strain along $\alpha = 45^\circ$. The misorientation of the Cu layer with respect to its initial orientation is obtained by measuring the angle between the x -direction, which initially coincides with the $[11\bar{1}]_{\text{Cu}}$ direction, and the current $[11\bar{1}]_{\text{Cu}}$ direction. As seen in Fig. 12, an initial and near constant misorientation of $\approx 10^\circ$ is measured. This is due in part to the glide of the Shockley partials from one Cu/Nb interface to another²⁰ and in part to the formation of nanotwins in the Cu layer. Remarkably, however, at larger strains, the misorientation suddenly increases and reaches a new plateau value. This misorientation jump correlates with the activation of additional slip systems in the Cu layer. At these large strains, however, the Cu orientation becomes relatively stable, which is

expected because deformation is then dominated by the localized shearing of the amorphous layers.

We also studied cyclic bi-planar shear in Cu–Nb for these same interfaces.³¹ In this case, multiple shearing directions are activated during each deformation cycle, leading to an averaging of the features reported here for single shear directions. Nevertheless, the main conclusions found for planar shear remain valid, as NW and {111}KS interfaces remain sharp up to equivalent strains of 50 and 150, respectively, while {112}KS interfaces undergo significant mixing, leading to the formation of amorphous layers at the original Cu–Nb interfaces.

CONCLUSIONS

The main conclusions of this article can be summarized as follows:

1. Chemical mixing forced by plastic deformation in two-phase alloys depends critically on the nature

of interfaces. For coherent or semicoherent interfaces, slip can transfer across interfaces and the resulting mixing displays an unusual dependence with the characteristic length scale of the microstructure. Simple geometric models and atomistic simulations show that this forced mixing is superdiffusive. For incoherent precipitates, however, dislocations pile up at interfaces, resulting in a diffusive forced mixing.

2. In the case of planar interfaces, as in multilayers, in addition to the above two mechanisms, interface dislocations may glide in the habit plane of the interface, leading to interface sliding and negligible chemical mixing.
3. These different mechanisms are illustrated by investigating three different interfaces in Cu-Nb multilayers, namely NW, {111}KS, and {112}KS. At strains up to several tens, the NW and {111}KS interfaces remain atomically sharp, and these multilayers deform by pure interface sliding, regardless of the shearing direction applied. Consequently, negligible mixing takes place at these interfaces.
4. In contrast, extensive mixing is observed when shearing the {112}KS interface along nearly all directions. The exception is when shearing proceeds along $\alpha = 90^\circ/270^\circ$, in which case interface sliding dominates at low strains. At larger strains, however, chemical mixing develops, and the mixing rate is similar to the ones measured for the other shearing directions.
5. For the {112}KS interface, chemical mixing leads to the formation of disordered, amorphous layers at the original Cu/Nb interfaces. At large strains, the imposed macroscopic deformation is accommodated by localized shearing of these amorphous layers. The amorphous layer thickness grows roughly as the square root of strain, which is consistent with a diffusive mechanism.
6. For the {112}KS interface, the crystallographic orientation of the Cu phase is roughly constant when deformation is dominated by localized shearing of the amorphous layers, but occasional bursts of dislocation activity result in jumps of the Cu crystal orientation, thus leading to respective orientations of the Cu and Nb phases far different from the initial ones imposed by the {112}KS orientation relationship.

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